Lab5

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(1)

Set $A_{TM} = \{<M, w> \mid M$ is a TM and M accepts $w\}$

$HALT_{TM} = \{<M, w> \mid M$ is a TM and M halts on $w\}$

Assuming $HALT_{TM}$ is decidable and a TM R decides $HALT_{TM}$.

Using R to construct S, for input $<M, w>$:

① If R rejects the input or R can not halt forever, S rejects the input.

② If R accepts the input, S accepts the input.

So S can decides A_{TM} and S is decidable.

According to A_{TM} is undecidable.

So $HALT_{TM}$ is undecidable.

(2)

① Since A is decidable, there exists a TM M that halts on every input w and M accept $w \Leftrightarrow w \in A$.

According to closure principle, M is a recognizer for A and $\bar{A}$.

If $w \in A$, M accepts; $w \notin A$, M reject.

If $w \notin A$, M' accepts; $w \in A$, M' accept.

So A and $\bar{A}$ are Turing-recognizable.

② A and $\bar{A}$ are Turing-recognizable.

Let TM M_1 can recognize A, TM M_2 can recognize $\bar{A}$.

Construct a TM M as follows:

M on input w:

(I) run M_1, M_2 in parallel on w:

Simulate one step of M_1 on one tape,

Simulate one step of M_2 on one tape.

(II) M_1 acepts, M accepts w.

M_2 accepts, M rejects w.

So it must halt.

M is a decider for A.

So A is decidable iff it is Turing-recognizable and co-Turing-recognizable.

(3) Language A is mapping reducible to language B, written $A \leq_m B$, if there is a computable function f: $\Sigma^* \to \Sigma^*$, where for every w, $w \in A \Leftrightarrow f(w) \in B$.
The function f is called the reduction of A to B.

(4) $A_{TM} \leq_m HALT_{TM}$

$A_{TM} = \{\langle M, w\rangle \mid M \text{ accepts } w\}$ $HALT_{TM} = \{\langle M, w\rangle \mid M \text{ halts on } w\}$

Defining a computable function f: $\langle M, w\rangle \in A_{TM} \Leftrightarrow f(\langle M, w\rangle) \in HALT_{TM}$

Let F be a TM that computes f.

F = On input $\langle M, w\rangle$:

Constructing TM M'.

M' = On input x:

① Run M on x

② If M accepts, M' accepts

③ If M rejects, M' goes to loop.

Output $\langle M', w\rangle$

So if M accepts w, M' accepts and halts, $\langle M', w\rangle \in HALT_{TM}$

If M rejects w or go to loop, M' enters a loop. $\langle M', w\rangle \notin HALT_{TM}$

So $A_{TM} \leq_m HALT_{TM}$

(5)

(a) M be a LBA with q states, g symbols in the tape alphabet and tape length n.

For current states, there exists q possible choices.

The tape has n position with possible g choices.

And q, g, n are independent.

Total configuration = (Number of state) * (Number of head positions) * (Number of tape contents)

$= q * n * g^n$

(b) $A_{LBA} = \{\langle M, w\rangle \mid M \text{ is a LBA that accepts input } w\}$

According to (a), the number of distinct configurations of M on a tape of length n is at most: $q * n * g^n$. If M runs for more than N steps, it must have repeated a configuration and enter an infinite loop.

Decider L:

L = On input $\langle M, w\rangle$, where M is a LBA:

① Comput $N = q * n * g^n$

② Simmulate M on w for at most N steps:

If M acepts within N steps, L accepts.

If M rejects within N steps, L rejects.

If M does not halt within N steps, L rejects.

So it can decide A_{LBA}.

A_{LBA} is decidable.